ML for Mech Eng (CoEP)

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Outline

About Me

Yogesh Haribhau Kulkarni

Bio:

- 20+ years in CAD/Engineering software development
- Got Bachelors, Masters and Doctoral degrees in Mechanical Engineering (specialization: Geometric Modeling Algorithms).
- Currently doing Coaching in fields such as Data Science, Artificial Intelligence Machine-Deep Learning (ML/DL) and Natural Language Processing (NLP).
- Feel free to follow me at:
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Office Hours: Saturdays, 2 to 5pm (IST); Free-Open to all; email for appointment.

Feature Extraction and Selection

From Raw Data to Useful Features

- Before training any model: which numbers do you feed it, and how many?
- **Feature Extraction**: raw data $\rightarrow$ a manageable set of numeric features.
- **Feature Selection**: given candidate features, which ones actually help?

Feature Extraction: Statistical Features

- Real signals often arrive as a stream: a sensor reading, a week of sales, a semester of test scores.
- A model needs one row per example, not 1000 raw readings – so summarize the stream instead.
- Simplest, most common summaries: mean, standard deviation, minimum, maximum, range.

Intuition

Describing a month of temperatures in one sentence? You wouldn't list 30 numbers – "averaged 22°C, swung between 18 and 27". That sentence *is* mean, min, max: statistical features.

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Feature Extraction: A Worked Example

A sensor logs one reading a minute, for 10 minutes:

```python
import numpy as np

readings = np.array([21.1, 21.3, 21.0, 21.6, 22.0,
                     22.4, 22.1, 21.8, 21.5, 21.2])

features = {
    'mean':  readings.mean(),
    'std':   readings.std(),
    'min':   readings.min(),
    'max':   readings.max(),
    'range': readings.max() - readings.min(),
}
print(features)

# Output:
# {'mean': 21.6, 'std': 0.44, 'min': 21.0, 'max': 22.4, 'range': 1.4}
```

10 numbers became 5 features. Per sensor, per time window – several sensors, dozens of candidates.

Feature Selection: Which Features Actually Help?

Feature Selection: Ranking (Filter Methods)

- Score each feature *on its own* against the target, rank, keep the top few.
- "Score": a statistical test, e.g. an F-test (ANOVA) for a numeric feature vs. a class label.
- Fast – one pass, no model training. Called a **filter**: features filtered before any model sees them.
- Blind spot: scores features in isolation, can miss a feature that only helps *in combination*.

Ranking: Loading the Data

Pima Indians Diabetes dataset: 768 patients, 8 measurements, one outcome.

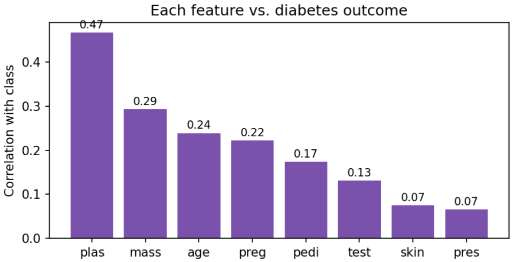
```
import pandas as pd

url =
    "https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.c
names = ['preg', 'plas', 'pres', 'skin', 'test', 'mass', 'pedi', 'age',
    'class']
df = pd.read_csv(url, names=names)
df.head(3)

# Output:
#    preg  plas  pres  skin  test  mass   pedi  age  class
# 0     6   148    72    35     0  33.6  0.627   50      1
# 1     1    85    66    29     0  26.6  0.351   31      0
# 2     8   183    64     0     0  23.3  0.672   32      1
```

class: 1 = diabetic, 0 = not. Which of the 8 columns actually predicts it?

Ranking: Correlation with the Outcome

Simplest ranking: correlation of each column with `class`.

```
corr = df.corr()['class'].drop('class').sort_values(ascending=False)
print(corr.round(2))
```



Ranking: F-score (a Sharper Test)

Correlation is one lens; an F-test (ANOVA) is another, built for exactly this:

```
from sklearn.feature_selection import SelectKBest, f_classif

X, y = df.values[:, 0:8], df.values[:, 8]
skb = SelectKBest(score_func=f_classif, k=4).fit(X, y)
ranked = sorted(zip(names[:8], skb.scores_), key=lambda t: -t[1])

# F-score, highest to lowest:
# plas 213  mass 72  age 46  preg 40
# pedi  24  test 13  skin  4  pres  3
```

Same story as the correlation plot: `plas` dominates, `skin`/`pres` barely register.

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Feature Selection: Wrapper Methods

- Filter scores each feature alone. **Wrapper**: train/cross-validate the model on different feature *subsets*, let that score decide.
- Slower – many models trained – but catches combinations a filter would miss.
- Which subsets to try? Four classic strategies:
  - Exhaustive Search
  - Greedy Forward Selection
  - Greedy Backward Elimination
  - Best-First Search

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Wrapper: Exhaustive Search

- Try *every* non-empty subset, keep the best cross-validated score.
- n features $\rightarrow 2^n - 1$ subsets. Pima's 8 $\rightarrow$ 255 – small enough to actually run.

```python
import itertools
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import cross_val_score, KFold

model = LogisticRegression(max_iter=1000)
kfold = KFold(n_splits=5, shuffle=True, random_state=7)
best_score, best_subset = -1, None

for r in range(1, 9):
    for combo in itertools.combinations(range(8), r):
        s = cross_val_score(model, X[:, combo], y, cv=kfold).mean()
        if s > best_score:
            best_score, best_subset = s, combo

# Output (255 subsets evaluated):
# Best: all features except 'skin', accuracy=0.775
# All 8 features:                    accuracy=0.775
```

Proves skin adds nothing: dropping it costs zero accuracy.

Wrapper: Why Exhaustive Search Doesn't Scale

- 8 features $\rightarrow$ 255 subsets: a few seconds.
- 3 sensors $\times$ 5 statistical features each $=$ 15-20 candidates already.
- 20 features $\rightarrow 2^{20} - 1 \approx 1$ million subsets – roughly 4,000$\times$ more work.
- Realistic only for a small, already-narrowed feature set.

Wrapper: Greedy Forward Selection

- Start empty. Each round: add whichever remaining feature helps most.
- Stop when nothing left improves the score.

Same model and cross validation, on Pima:

Step	Feature added	Accuracy so far
1	plas	0.750
2	mass	0.767
3	pedi	0.773
4	pres	0.772

3 features reach 0.773 – within 0.002 of exhaustive search's best, at a fraction of the cost.

Wrapper: Greedy Backward Elimination

- Start with *all* features. Each round: remove whichever hurts least (or helps).
- Stop once every remaining removal costs accuracy.

On Pima:

Step	Feature removed	Accuracy after
start	(all 8)	0.775
1	skin	0.775
2	age	0.772
3	preg	0.775

First move – drop `skin` – already matches what exhaustive search proved was right.

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Wrapper: Best-First Search

- Greedy commits forever: once added or removed, never reconsidered.
- Best-first keeps a pool of promising subsets, not just one – can expand any of them next.
- Can back out of an early choice greedy would be stuck with.
- Costs more than greedy, far less than exhaustive.

Intuition

Greedy: always take the steepest visible path uphill. Best-first: keep a mental map of your 3 best viewpoints, and choose which to climb from next.

Filter vs. Wrapper: Summary

Method	Looks at	Cost
Ranking (Filter)	One feature at a time	Cheapest
Exhaustive	Every possible subset	Guaranteed best, but explodes
Greedy Forward	Builds up, one add at a time	Cheap, no backtracking
Greedy Backward	Trims down, one remove at a time	Cheap, no backtracking
Best-First	A pool of candidate subsets	Middle ground

Intuition

No method is "correct." Ranking: fast first pass. Wrapper: costs more, sees combinations. Pick by budget.

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Quick Check: Feature Selection

Think About It

The filter ranking on Pima placed `preg` (number of pregnancies) 4th out of 8 individually. Yet Greedy Backward Elimination's best 5-feature subset dropped `preg` entirely without hurting accuracy. Are these two results in conflict? Why or why not?

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Quick Check: Feature Selection

Answer

No conflict – they answer different questions. The filter score measures how strongly `preg` relates to the outcome *by itself*, ignoring every other feature; on that measure it ranks 4th. Backward elimination asks a different question: given everything else already in the model, does `preg` add anything *on top of it*? If the information `preg` carries is already captured by other features (e.g. age correlates with number of pregnancies), removing it costs nothing even though it looked useful alone. This is exactly the wrapper methods' advantage over filters: they can see redundancy that isolated ranking cannot.

Recap: Feature Extraction and Selection

- **Extraction**: raw data $\rightarrow$ manageable features (mean, std, min, max, range).
- **Selection**: which candidate features earn a place in the model.
- **Filter (Ranking)**: fast, scores features independently.
- **Wrapper** (Exhaustive, Greedy Forward/Backward, Best-First): slower, sees combinations filters miss.
- Narrows the search – you still decide what ships.

Hyperparameter Tuning

Hyperparameter Tuning

- **Parameters**: learned from data during `fit()` – e.g. regression coefficients, a tree's split points.
- **Hyperparameters**: set *before* training, control how learning happens – e.g. k in KNN, `max_depth` in a Decision Tree, C in an SVM.
- Picking by hand ("try $k = 3$, try $k = 5$, . . . ") doesn't scale past one hyperparameter.
- Fix: try many combinations, score each with cross validation, keep the best.

Intuition

Parameters: what the model learns. Hyperparameters: the knobs you set before it starts learning – oven temperature and bake time, chosen before the cake goes in.

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Hyperparameter Tuning: Grid Search

Try every combination on a fixed grid, keep the best cross-validated score.

```
import pandas as pd
from sklearn.model_selection import GridSearchCV
from sklearn.neighbors import KNeighborsClassifier

url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.c
names = ['preg', 'plas', 'pres', 'skin', 'test', 'mass', 'pedi', 'age', 'class']
df = pd.read_csv(url, names=names)
X, y = df.values[:, 0:8], df.values[:, 8]

model = KNeighborsClassifier()
param_grid = {'n_neighbors': [3, 5, 7, 9, 11, 15, 21]}

grid = GridSearchCV(model, param_grid, cv=5, scoring='accuracy')
grid.fit(X, y)

# Output: Best k 11, accuracy 0.749
```

Hyperparameter Tuning: Grid Search, Cost

- 7 candidate values of $k \times 5$ folds $= 35$ model fits.
- All run automatically – no manual loop.
- One hyperparameter is cheap. Cost multiplies fast with more: 3 hyperparameters, 10 values each, $= 1000$ fits.

Hyperparameter Tuning: Randomized Search

- Grid Search checks every combination – fine for one hyperparameter, explodes with more.
- `RandomizedSearchCV`: sample a fixed number of random combinations instead.
- Trades a guarantee of the single best combination for a large cut in compute.

```python
from sklearn.model_selection import RandomizedSearchCV

param_dist = {'n_neighbors': list(range(1, 30))}
random_search = RandomizedSearchCV(model, param_dist, n_iter=5,
                                   cv=5, scoring='accuracy', random_state=42)
random_search.fit(X, y)

# Output: tried 5 of 29 possible values -> Best k 13, accuracy 0.755
```

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Hyperparameter Tuning: Randomized Search, Result

- Just 5 candidates out of 29 possible values of k.
- Matched Grid Search's result ($k = 11$ vs. $k = 13$, 0.749 vs. 0.755) at a fraction of the cost.
- In practice: often close to Grid Search's answer, rarely worse in any way that matters.

Quick Check: Hyperparameter Tuning

Think About It

Why is it important to use cross-validated scores, not a single train/test split, when comparing hyperparameter values during Grid or Randomized Search?

Quick Check: Hyperparameter Tuning

Answer

A single split can hand a particular hyperparameter value a "lucky" or "unlucky" test set by chance. Averaging over k folds gives a far more reliable estimate of how each candidate value actually generalizes, so the value selected as "best" is best on average, not best on one arbitrary split.

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Hands-On: End-to-End Mini Pipeline

The Dataset

- Steel Plates Faults dataset (UCI) – a real manufacturing quality-control problem.
- 1941 steel plates, 27 geometric/luminosity measurements each, one of 7 recorded defect types per plate.
- Task: detect `K_Scatch` – a scratch-type surface defect – vs. everything else.
- 27 features – too many to exhaustively search, a real test for ranking and wrapper methods.
- Goal: pandas to load and inspect, sklearn to preprocess/select/train/evaluate – a revision lap before the algorithm-by-algorithm sessions start.

Step 1: Load and Look

```python
import pandas as pd

feature_names = ['X_Minimum', 'X_Maximum', 'Y_Minimum', 'Y_Maximum',
    'Pixels_Areas', 'X_Perimeter', 'Y_Perimeter', 'Sum_of_Luminosity',
    ]  # ... 27 columns total, see dataset README
fault_names = ['Pastry', 'Z_Scratch', 'K_Scatch', 'Stains',
               'Dirtiness', 'Bumps', 'Other_Faults']

df = pd.read_csv('Faults.NNA', sep=r'\s+', header=None,
                 names=feature_names + fault_names)
df['target'] = df['K_Scatch']
print(df.shape)
df.head(3)

# Output: (1941, 35)
#    X_Minimum  X_Maximum  Y_Minimum  ...  target
# 0         42         50     270900  ...       0
# 1        645        651    2538079  ...       0
```

Step 2: Missing Values and Class Balance

```
print(df.isna().sum().sum(), "missing values")
print(df['target'].value_counts())

# Output:
# 0 missing values
# target
# 0    1550   (no K_Scatch defect)
# 1     391   (K_Scatch defect)
```

Step 3: Train/Test Split

Stratify on the target so both splits keep the same 80/20 class balance.

```python
from sklearn.model_selection import train_test_split

X = df[feature_names].values
y = df['target'].values

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y)

# Output: X_train (1455, 27)   X_test (486, 27)
```

Step 4: Scale the Features

Fit the scaler on training data only – the same no-leakage rule from Data Preparation.

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler().fit(X_train)
X_train_s = scaler.transform(X_train)
X_test_s  = scaler.transform(X_test)
```

Step 5: Baseline, All 27 Features

```python
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

model_all = LogisticRegression(max_iter=5000)
model_all.fit(X_train_s, y_train)
pred_all = model_all.predict(X_test_s)
print("Accuracy:", accuracy_score(y_test, pred_all))

# Output: Accuracy: 0.977
```

Step 6: Rank Features (Filter)

Same `SelectKBest` approach as before, now on 27 candidates instead of 8.

```python
from sklearn.feature_selection import SelectKBest, f_classif

skb = SelectKBest(score_func=f_classif, k=10).fit(X_train_s, y_train)
top10 = [feature_names[i] for i in skb.get_support(indices=True)]
print(top10)

# Output: ['Pixels_Areas', 'X_Perimeter', 'Sum_of_Luminosity',
#          'Minimum_of_Luminosity', 'Outside_X_Index', 'Edges_Y_Index',
#          'LogOfAreas', 'Log_X_Index', 'Log_Y_Index', 'SigmoidOfAreas']
```

Step 7: Retrain on the Top 10

```python
idx = skb.get_support(indices=True)
model_10 = LogisticRegression(max_iter=5000)
model_10.fit(X_train_s[:, idx], y_train)
pred_10 = model_10.predict(X_test_s[:, idx])
print("Accuracy:", accuracy_score(y_test, pred_10))

# Output: Accuracy: 0.951
```

Step 8: Filter Result – Not a Free Lunch

- 27 features: 0.977 test accuracy.
- Filter top 10: 0.951 test accuracy.
- 17 dropped features weren't dead weight – accuracy fell by $\sim$2.6 points.
- Real trade-off: fewer features for a real accuracy cost. Worth it or not depends on the deployment.

Step 9: Try a Wrapper Instead

Greedy Forward Selection, cross-validated, on all 27 features:

```python
from sklearn.model_selection import cross_val_score, KFold

Xs = StandardScaler().fit_transform(X)
kfold = KFold(n_splits=5, shuffle=True, random_state=7)
selected, remaining = [], list(range(27))

for step in range(6):
    scores = [(f, cross_val_score(LogisticRegression(max_iter=5000),
                    Xs[:, selected + [f]], y, cv=kfold).mean()) for f in remaining]
    best_f, best_s = max(scores, key=lambda t: t[1])
    selected.append(best_f); remaining.remove(best_f)

# Output: 6 rounds -> Log_X_Index, Steel_Plate_Thickness, Pixels_Areas,
#         Square_Index, X_Minimum, Orientation_Index
```

Step 10: Filter vs. Wrapper vs. Everything

Feature set	# Features	5-fold CV Accuracy
All features	27	0.975
Filter (top by F-score)	10	0.949
Wrapper (greedy forward)	6	0.963

Intuition

6 wrapper-chosen features close most of the gap to all 27 (0.963 vs. 0.975) while clearly beating the 10-feature filter (0.963 vs. 0.949) – fewer, better-combined features can rival throwing everything at the model. The filter's top 10, chosen one at a time, missed a combination the wrapper found.

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Step 11: Evaluate the Final Model

```python
from sklearn.metrics import confusion_matrix, classification_report

wrapper_cols = ['Log_X_Index', 'Steel_Plate_Thickness', 'Pixels_Areas',
                'Square_Index', 'X_Minimum', 'Orientation_Index']
idx6 = [feature_names.index(f) for f in wrapper_cols]

model_6 = LogisticRegression(max_iter=5000)
model_6.fit(X_train_s[:, idx6], y_train)
pred_6 = model_6.predict(X_test_s[:, idx6])
print(confusion_matrix(y_test, pred_6))

# Output: [[384   4]
#          [ 16  82]]
```

Step 11: Classification Report

```
print(classification_report(y_test, pred_6,
                            target_names=['no_k_scatch', 'k_scatch']))

# Output:
#                precision  recall  f1-score  support
#  no_k_scatch       0.96    0.99      0.97      388
#     k_scatch       0.95    0.84      0.89       98
```

6 features, 96% test accuracy, beating both the 10-feature filter model and coming close to the full 27-feature model – on cross validation too, not just this one split.

Quick Check: The Pipeline

Think About It

Step 6 fits `SelectKBest` on `X_train_s` (the scaled training set), not the full scaled dataset. Step 9's greedy forward selection, however, scores each candidate using cross validation on *all* of `X` (via `Xs`, not `X_train_s`). Is Step 9 leaking test information in a way Step 6 avoids?

Quick Check: The Pipeline

Answer

No – both are safe, but for different reasons. Step 6 never touches `X_test_s` at all. Step 9 does use every row of `X`, including what would later become test rows, but cross validation itself holds out a fold at a time during scoring – so no fold's held-out data influences its own score. The real risk it does *not* avoid: the final held-out test set (`X_test_s`) from Step 3 was never part of Step 9's cross validation either, so the eventual Step 11 evaluation is still a fair, unseen check on the wrapper-selected features.

Try It Yourself

- Change `k=10` to `k=5` in Step 6 – does accuracy hold up with even fewer features?
- Run Greedy Backward Elimination instead of Forward on all 27 features – same 6 features at the end?
- Swap `LogisticRegression` for a different classifier – does the winning feature set change?
- Try a different target fault type (e.g. `Bumps`) – does the same feature set still win? (Spoiler: no – worth discussing why.)

Recap: The Pipeline Pattern

- Load → Inspect → Split → Scale → Select → Train → Evaluate → Compare.
- Every step reused today: pandas for loading/inspecting, sklearn for the rest.
- This shape doesn't change once you swap in a real ML algorithm – only Step 5/7's model does.
- Next: the algorithms themselves, one at a time, starting with Linear Regression.

Thanks . . .

- Office Hours: Saturdays, 3 to 5 pm (IST); Free-Open to all; email for appointment to yogeshkulkarni at yahoo dot com
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